

# Additive Manufacturing of an Anthropomorphic Hand using Force Resistive Sensors for Object Detection

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## Abstract

Anthropomorphic hands have become increasingly popular for applications in recent decades. They are used in the development of prosthetics, surgical robots, and teleoperation. However, current robotic hands are limited in their grip force sensitivity and expense. A clinically prescribable and commercially available (CPCA) prosthetic can cost upwards of \$20,000 USD. In order to develop a low cost and high performance anthropomorphic hand, we used multimaterial 3D printing with attached low cost film pressure sensors. We used PLA (Polylactic Acid) for the fingers and arm and TPU (Thermoplastic Polyurethane) for the joints and palm, and integrated force resistive sensors to detect pressure when grabbing objects. Training data was collected by grabbing four unique objects, then a random forest machine learning algorithm was used to identify the objects based on sensor inputs. The model sustained an accuracy of 98.1% from 4500 rows of training data and 1000 rows of testing data. We supported our hypothesis believing additive manufacturing can be combined with force resistive sensors to create a low cost bioinspired hand with object detection capabilities. Accurate results in object detection may be expanded on with potential applications in the advancement of adaptive grasp, soft robotics, smart prosthetics, AI, human skin biomimicry, and the consideration of material properties in computer vision.

## Introduction

The human hand is considered one of the most dextrous manipulators on Earth, with the range to grasp or crush almost anything [12]. Giving human characteristics and appearances, like the shape, structure, and ability of the human hand, to inanimate objects, is called anthropomorphism [18]. Within the invention of robotic hands for use in prosthetics, surgery, and teleoperation, the necessity of the ability of robotic hands to explore and modify the environment like a human hand increases [7]. Including the integration of human hand characteristics into the design of robotic hands, the dexterity and precision of grabbing can be improved. Current uses of anthropomorphic robotic hands include: open-source 3D printed hands like the HACKberry Hand, the HANDi Hand, and the BEAR PAW [17] as prosthetics, robotic hand and arm

structures for precise and minimally invasive surgeries [20], and teloperation, where robotics hands and arms are controlled by humans [3]. High quality CPCA prosthetic devices can cost more than \$20,000 USD [17]. Current limitations in 3D printed hands compared to their machined counterparts include: inadequate grip strength, insufficient friction, and design factors that prevent the effective reorientation of contact forces [17]. Choosing a more cost-effective motor or controller can decrease hand functionality. Balancing cost with quality of product, by use of 3D printing can decrease expense of product.

Additive manufacturing, also referred to as 3D printing, has rapidly grown into a widely used technology. The process involves the physical creation of a digital three-dimensional model created by Computer-Aided Design (CAD) through the addition of material layer by layer [16]. There are many types of 3D printing including Stereolithography (SLA) and Binder Jetting, but the most common type of 3D printing, which was used for our anthropomorphic hand, is Fused Deposition Modeling (FDM). FDM creates objects by feeding material, in the form of filament, into an extruder, then heating it up and layering it down on a bed [19].

The advancement of 3D printing over the years is largely due to its multitude of benefits. Most importantly, 3D printing is both cost effective and fast, transforming rapid prototyping [9]. Beyond that, it also allows the creation of more complex shapes without additional cost, increasing design customization [1]. These advantages have led 3D printing to be applied to a variety of industries. It is used to create lightweight and affordable parts in the aerospace and automotive industries or to design easily customized parts for electronics or to more efficiently print environmentally friendly buildings. 3D printing is also used to create skin, tissues, bone, organs, and cartilage for the healthcare and medical industries. Furthermore, it has even been implemented in the fashion and food industries, allowing for 3D printed shoes, jewelry, and clothes as well as customizable chocolate, pasta, and pizza [16].

Cost effective anthropomorphic hands are built with 3D printing for many reasons. First of all, 3D printing is affordable and efficient when making many design iterations. Since it also allows for the creation of detailed designs, it is well suited for the creation of the complex structure of a human hand. The joints in the fingers are the fundamental part of the hand, and 3D printing allows for a multitude of different joints that can mimic natural human joints [14]. In addition to this, 3D printing also offers a variety of materials that can range from commercial levels, such as Polylactic acid (PLA) and Thermoplastic Polyurethane (TPU) to industrial-grade metals, ceramics, polymers, and composites.

Robotic hands are inspired by human hands because of their efficiency and versatility. The unique structure and movement of the human hand, with bones, joints, muscles, tendons, and nerves, allows for precise function. For example, robotic fingers contain realistic joints, while strings and motors mimic muscle and tendon movement [8,15]. This hand contains all distal interphalangeal, proximal interphalangeal, and metacarpophalangeal joints. 3D printing

allowed the hand to be customizable, so we based the size and shape of ours on our team member Christopher Huang's hand. This bioinspiration allows for the robotic hand to mimic the dexterity of a human hand's geometry. It can complete complex tasks by copying the flexibility and adaptability of human hands. Robotic hands can grasp objects of different shapes, weights, and sizes without damage. This is useful in many fields including healthcare, manufacturing, and service [4,5]. Adapting components of the human hand also helps robotic hands have more natural and efficient control systems [8]. The use of sensors can aid robotic hands in reaction speed and accuracy by detecting contact patterns. Being able to sense pressure mimics the function of mechanoreceptors in the skin, allowing the robot hand to respond like humans sensing touch [6,21]. Additionally, we have considered artificial skin with fingerprint ridges and skin creases modelled after existing applications of tactile sensing electronic skin [10]. The texture will help the robotic hand grip with less strength and more precision [2].

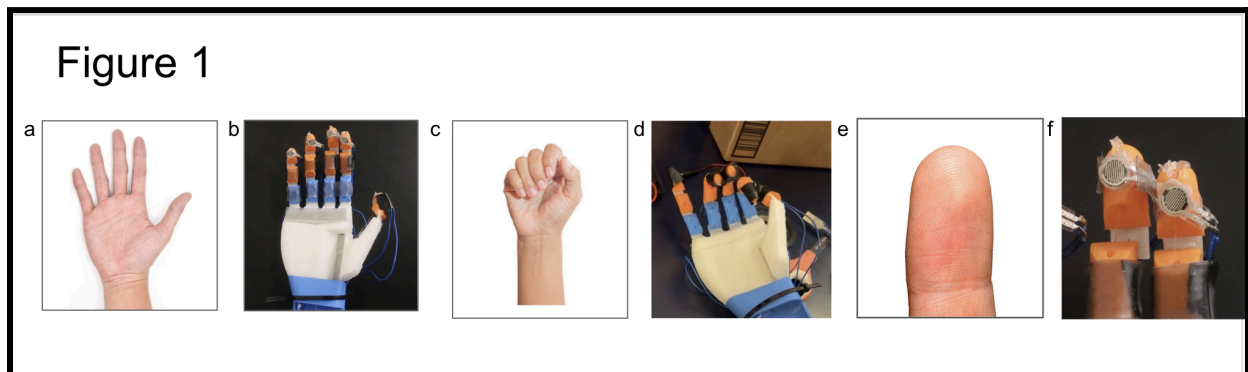


Figure 1: The bioinspiration in shape, structure, and size from a human hand, as well as the similarities in hand function and movement. “a” and “b” show the likeness between the size and shape of a human hand and the robotic hand, “c” and “d” show the similarities in hand movement, and “e” and “f” show the correspondence in hand function.

To begin the creation of the robotic hand, inspired by a real human hand as shown in Figure 1, the parts were designed through CAD and then 3D printed. 3D printing is a popular method for simple prototyping, testing, and improving, due to its quick manufacturing time and inexpensive costs. Different materials were used for different functions and areas of the hand. The combination of rigid and flexible materials creates a hand that is durable and adaptable. The hand has grabbing strength similar to a normal hand and can hold objects of various shapes. This simple claw contains force resistive sensors on the fingertips that produce feedback on pressure readings from objects (see Figure 1f). An electronic system in and on the forearm of the hand contains sensors for detection and motors for grasping. Therefore, the robotic hand can grab different objects and collect data via the sensors (see Figure 1b and 1d). We used this data to train a Random Forest machine learning model for object identification. This model is highly effective for classification tasks where it makes decisions based on observed previous patterns. The Random Forest machine learning model reduces error through many decision trees, works

relatively well with smaller data sets, and is able to handle sensor fluctuations. Random Forest models are very accurate and easy to train which allows for applications that can easily use object detection to readjust grip strength or perform further instructions. This anthropomorphic hand costs about \$103.07 to produce, giving it viability for low cost tactile sensing, and the object detection model makes it applicable in fields like agriculture, cooking, research, and computer vision.

## Materials and Methods

All components were created using the computer-aided design (CAD) softwares Onshape and Fusion 360. The models were then exported as a stereolithography (STL) file and imported into the slicing software Bambu Studio. Fused deposition modeling (FDM) parts were printed on a Bambu Lab P1S (build volume: 256 x 256 x 256 mm) and a Bambu Lab A1 mini (build volume: 180 x 180 x 180 mm). A 0.4 mm nozzle was used for all parts and PLA components were printed with a nozzle temperature of 220 °C, while TPU components were printed at 240 °C. The overall size of the assembled model was 466 mm in length and 149 mm in width. Individual PLA fingers ranged from 29mm in length for the thumb to 97mm for the middle finger. The TPU finger joints measured 8mm in length, while the TPU palm measures at 107mm vertically. The wrist/forearm piece measures at 239mm in length. The wheels vary in diameter from 32mm for the thumb to 35mm for the pointer finger. Print settings were optimized based on the function of each component. PLA fingers and wheel components were printed with a grid pattern and 15% infill. TPU joints were also printed with a 15% grid infill pattern. The TPU palm was printed with a 6% gyroid infill pattern. The wrist and forearm were printed with an infill of 8%. After FDM printing, all parts went through post processing, including removing supports and surface cleaning as needed. The fingertip caps were printed with EL 400 Resin on a Phrozen Sonic Mini 8K stereolithography (SLA) printer. The files were prepared for printing using the slicing software Chitubox. After printing, the parts were removed off the build plate, washed in a cleaning solution for 10 minutes and cured with UV light for between 5 and 10 minutes.

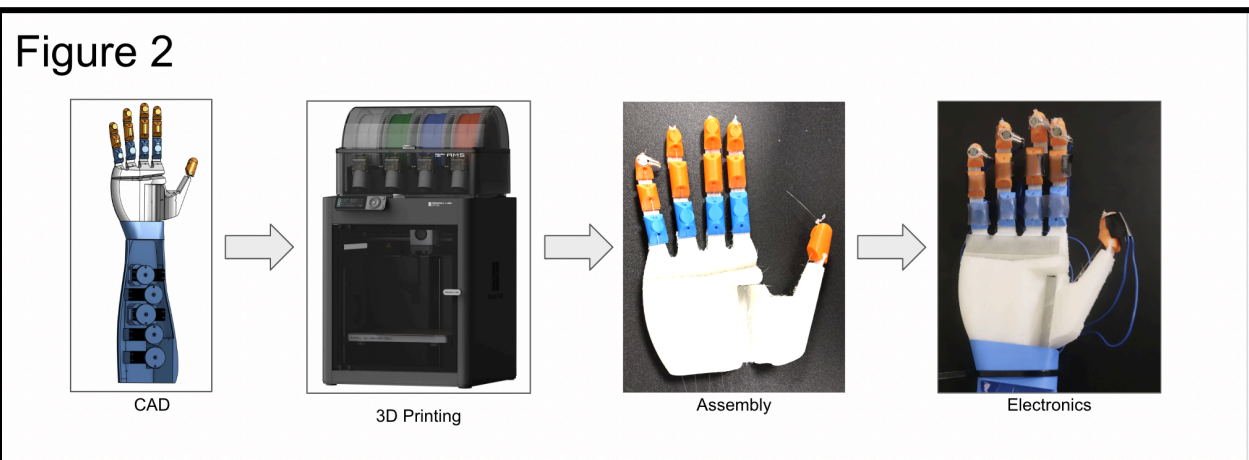


Figure 2: The flow chart of the design and assembly of the anthropomorphic hand. First, the whole hand was modeled in CAD, then 3D printed on a Bambu Lab P1S or A1 mini. Then it was assembled, first by the fingers, which were then added to the palm and strung with fishing line so the fingers would contract. Then it was attached to the wrist and forearm, and the electronics were soldered and attached.

As shown in Figure 3, the hand is controlled by an Arduino Uno, which can be powered by a nine-volt battery or through the connection to a computer. A 6V Tenergy battery pack powers a pin board, which has five Miuzei servos and five USDRWAM pressure sensors connected with unleaded solder. The pinboard was made by first soldering in a positive power rail and a negative power rail from the Tenergy battery pack, then by soldering the positive pressure sensor wires into the board, then to the power rail. The negative end of the pressure sensor was then soldered into the board in the same vertical row as the communication wire and the 10 kilo-ohm resistor, which were then all connected with solder to ensure a strong connection. The communication wire, which reads the voltage drop across the sensor and outputs it as a resistance value, was then plugged into the Analogue ports of the Arduino Uno. Finally, the other end of the resistor was soldered to the negative power rail of the board. For the servos, the positive power wire was soldered first into the board, and then to the power rail, with the communications wire, which delivers instructions to the motor, skipping the pin board entirely and just plugging into the PWM (Pulse Width Modulation) ports on the Arduino Uno, designated with a ~ in front of the number. Then the negative wire was soldered to the pin board and then to the negative power rail.

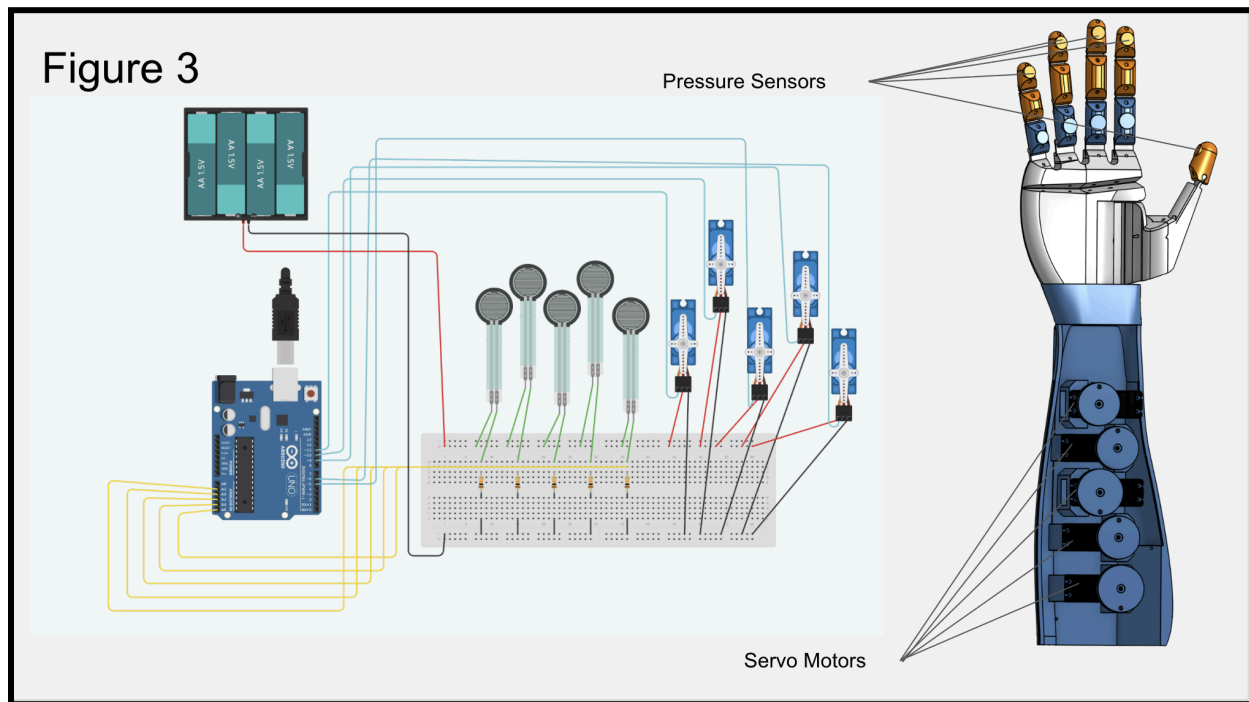


Figure 3: The wiring diagram and setup of the robotic hand, with the 6V battery pack powering the pinboard with the pressure sensors and servos, as well as where the pressure sensors and servos connect on the hand.

The robotic hand was tested through force resistive sensor data after grabbing four different objects. A chip bag weighing 31.7641 grams, a rice krispy treat weighing 22.1126 grams, a chip box weighing 25.5187 grams, and a Sharpie box weighing 15.9577 grams. In testing, the hand was held from the forearm slanted facing outward. The object was held against the hand mimicking natural actions while maximizing sensor surface area in the grip. After running code to grab the object, we let go so only the robot hand is holding it and begin data collection from the sensors. The data was saved in Excel as one set of inputs from every finger every 0.1 seconds. The grip was the same each test, and multiple tests were conducted for a total of around 5500 lines of data. The data measures resistance which is positively correlated with the force of grabbing the object.

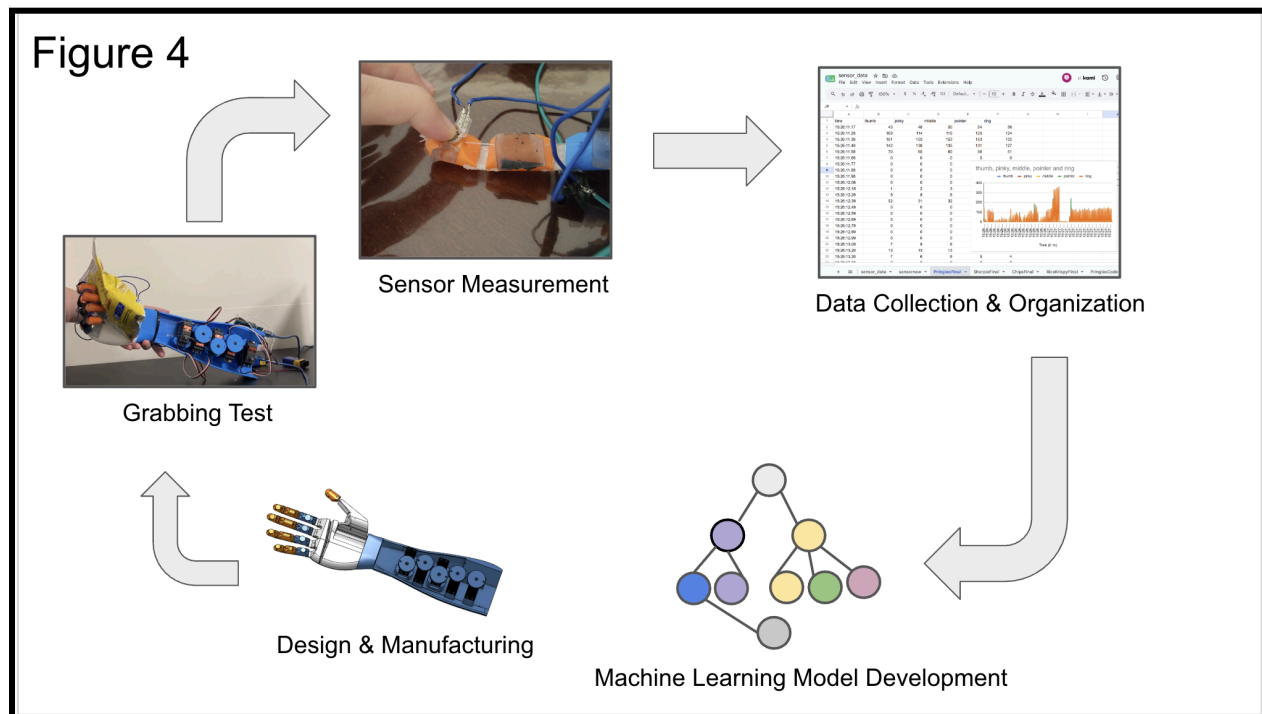


Figure 4: The flow chart that led to the training and development of the Random Forest Machine Learning Model. We started with the design and manufacturing of the anthropomorphic hand, which led to a grabbing test to ensure we could properly pick up the objects. The success with the grabbing test led to grabbing the objects with the pressure sensors running, which allowed us to collect and organize the data readings, and put them into a machine learning model to train it for object detection.

## Results and Discussion

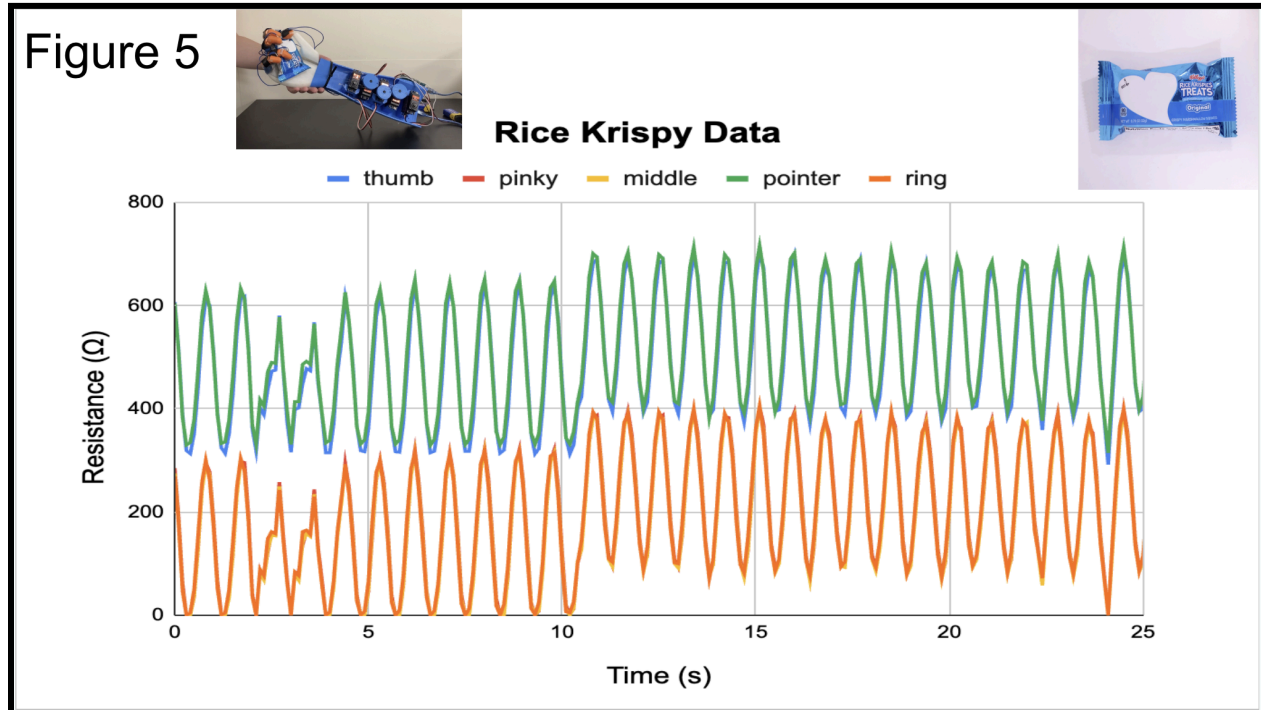


Figure 5: The Rice Krispy data, with the resistance values (in  $\Omega$ ) graphed by finger per second when grabbing a rice krispy bag.

Figure 5 shows that from 0 to 10 seconds, the thumb and pointer finger fluctuate, due to the natural fluctuations in the sensors we used, from  $300\Omega$  to slightly above  $600\Omega$ . In the same time frame, the ring, pinky, and middle fingers fluctuate from  $0\Omega$  to  $300\Omega$ . From the higher maximums of the thumb and pointer fingers, we can see that they have a tighter grip on the object, and thus exert more force on the object, which reads as a higher resistance level. From 10 to 25 seconds, the thumb and pointer fingers are still exerting the most force, but the numbers increase slightly, fluctuating from  $400\Omega$  to  $700\Omega$ . The ring, pinky and middle fingers resistance levels have also increased, fluctuating  $100\Omega$  to  $400\Omega$ . They are still exerting less force on the object than the thumb and pointer fingers, but their grip has tightened slightly.

Figure 6

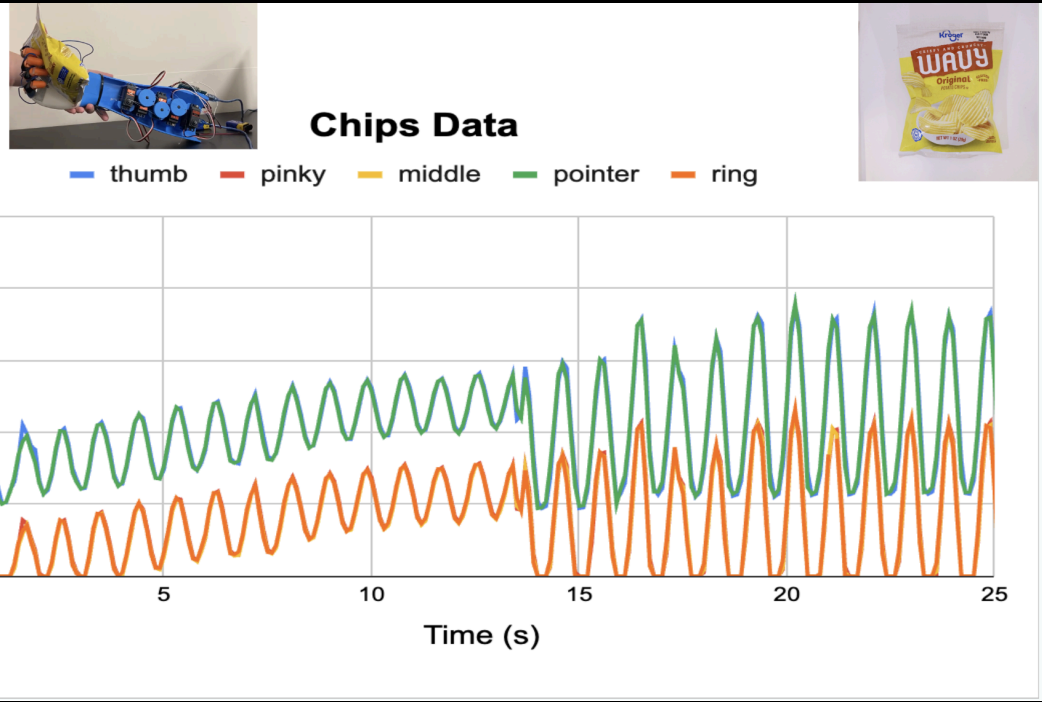


Figure 6. The chips data, with the resistance values (in  $\Omega$ ) graphed by finger per second when grabbing a chip bag.

Figure 6 data marks an increasing amount of resistance, with oscillations increasing from 0 to 14 seconds. On the pointer finger, the lowest resistance ranged from  $100\Omega$  to about  $170\Omega$  in the first 2 seconds, with the amplitude staying the same as the overall values increased from 2 to 14 seconds. On the ring finger, the resistance values ranged from  $0\Omega$  to  $80\Omega$  in the first 2 seconds, and also increased in overall value with the same amplitude from 2 to 14 seconds. After 14 seconds, the amplitude of the resistance oscillations increases dramatically, ranging from  $100\Omega$  to  $400\Omega$  on the pointer finger and  $0\Omega$  to  $250\Omega$  on the ring.

As the chip bag is grabbed for a longer amount of time by the hand, the resistance experienced by the pressure sensors also gradually increases. The shift in oscillation can be explained by a natural shift in the bag's position during the grabbing process, since it is a comparatively malleable object.

Figure 7

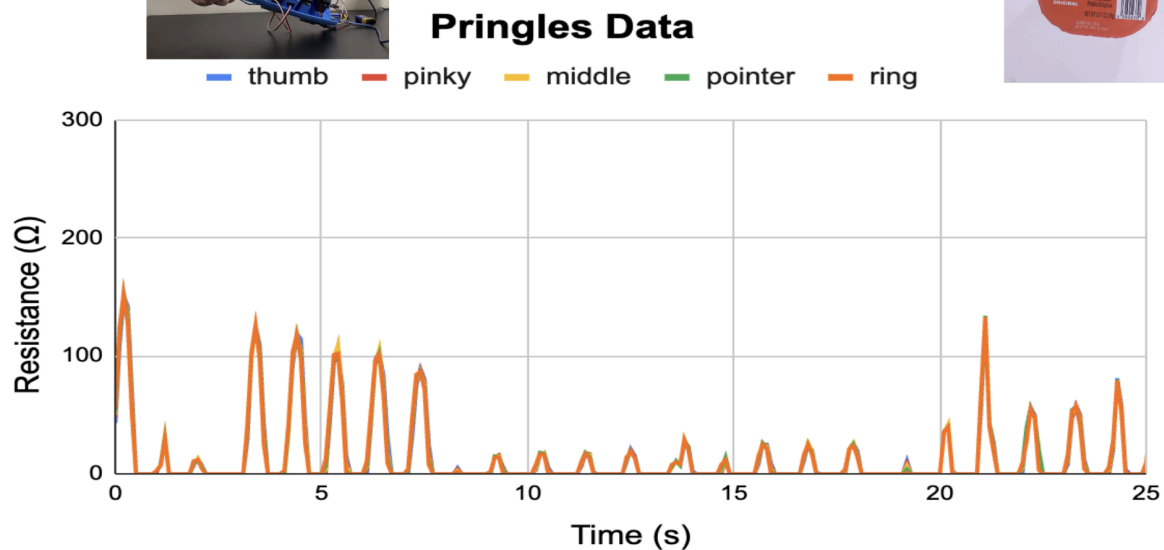


Figure 7: The Pringles Data, with the resistance values (in  $\Omega$ ) graphed by finger per second when grabbing a Pringle box.

Figure 7 shows that all fingers maintain a similar pressure throughout the entire trial so each finger gripped the object with the same amount of force. The pressure data at first rapidly decreases from 150 $\Omega$  to 90 $\Omega$  then slowly lowers until 10 seconds. Then from 10 to 20 seconds all fingers maintain low pressure around 20 ohms before rising to around 150 ohms at 21 seconds then dropping again at 22 seconds before gradually increasing until 25 seconds.

Figure 8

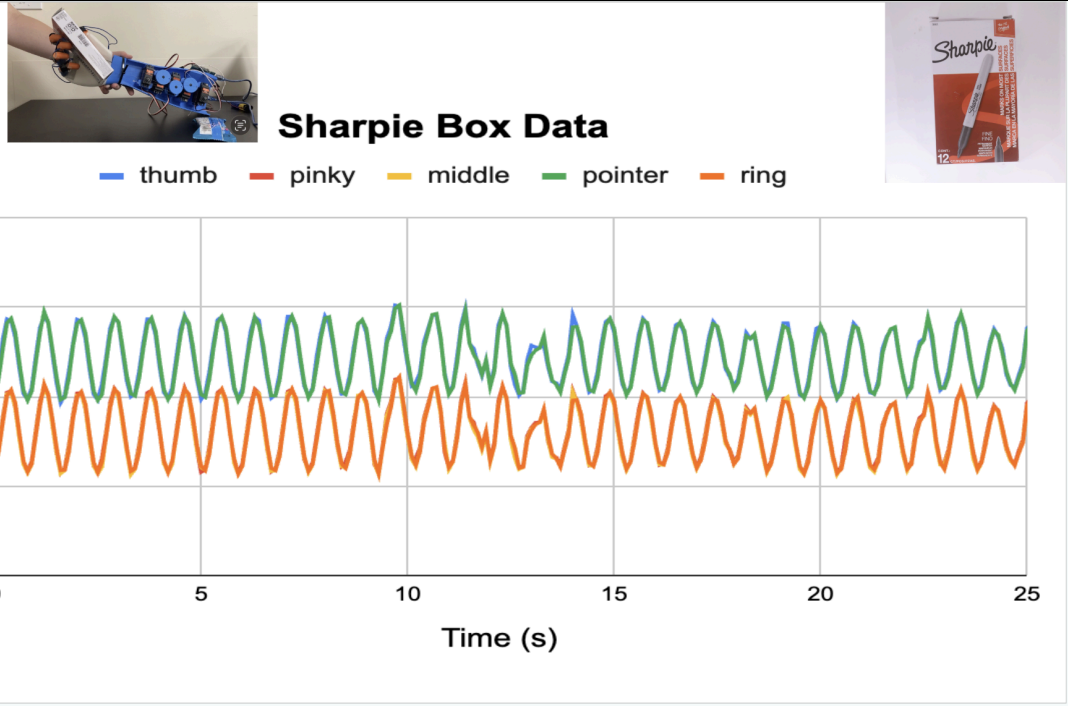


Figure 8. The Sharpie Box Data, with the resistance values (in  $\Omega$ ) graphed by fingers per second when grabbing a sharpie box.

Figure 8 shows the data remained consistent throughout the testing period as it rose and fell in a wave-like pattern every 0.8 seconds. There was a slight change from 10 seconds to 15 seconds, though, as the resistance for all the fingers lowered a bit before rising again after a while. There was also a spike for the thumb and middle finger when it rose again. The resistance values stayed between about 200 to 600 ohms throughout the whole data for each finger. The pinky, middle, and ring fingers stayed between about 200-400 ohms, while the pointer and thumb stayed between about 400-600 ohms. This suggests that the pointer and thumb have a tighter grip on the sharpie box compared to the rest of the fingers. We can also conclude that this data is different from the others because it does not show any major or significant change throughout the test, except between the 10-15 second interval, when there was a slight change in resistance values.

When grabbing a soft object such as the chips or pringles box, there was significantly less pressure applied compared to the rigid objects. Thus the resistance is smaller for less pressure compared with higher pressure. This is shown through the resistance data, which marks more rigid objects like the sharpie box and rice krispy with resistance values between  $0\Omega$  and  $400\Omega$  for the ring finger, while more malleable objects like the chips and pringles have ring finger resistance between  $0\Omega$  and  $200\Omega$ .

Table 1: Accuracy of Random Forest Model in Object Detection

Training Rows: 4546, Testing Rows: 1137

Accuracy: 98.1%

|              | Precision | Recall | f1-score | Support |
|--------------|-----------|--------|----------|---------|
| Box          | 0.98      | 0.95   | 0.96     | 278     |
| Chips        | 0.97      | 0.99   | 0.98     | 280     |
| Pringles     | 1.00      | 0.99   | 0.99     | 299     |
| Rice Krispy  | 0.98      | 0.99   | 0.98     | 280     |
| Accuracy     |           |        | 0.98     | 1137    |
| Macro Avg    | 0.98      | 0.98   | 0.98     | 1137    |
| Weighted Avg | 0.98      | 0.98   | 0.98     | 1137    |

Table 1 shows the accuracy of the Random Forest Model overall, at 98.1%, as well the accuracy of the model for each individual object. The f1-score uses a harmonic mean to consider both the precision of the model, examining how many of the model's guesses were correct, and its recall, examining how many of the correct options were guessed by the model. The support column states the number of occurrences of each object within the dataset. The pringles are the most accurate object with a f1-score of 0.99, because of the unique grip where the sensors barely touch the container. The other objects are also very accurate because they have different mechanical properties. The rice krispy has a f1-score of 0.98 and is hard originally, but deforms easily with constant pressure. The Sharpie box has a f1-score of 0.96 and is stiff but flexible with a larger shape. The chips bag has a f1-score of 0.98 and is soft and elastic, and is slightly less accurate because of its changing shape and shifting mass as it is grabbed. Overall, the machine learning model is extremely accurate in object detection because the objects have distinct grip patterns reflected through the sensor data.

This anthropomorphic hand expands on current designs by combining the foundation of a low cost 3D printed cable actuated hand with low cost pressure sensors. Together, these components allow for the development of pressure sensing hands that are suitable for object detection with extremely low manufacturing costs. The low cost of around \$103.07 makes the hand particularly viable for the research, prototyping, and distribution of new technologies. The

hand provides a highly accessible base for models like slip detection and adaptive grasp to be implemented into usable prosthetics and robotics that rely on low latency tactile sensing information.

The development of an object detection machine learning model demonstrates possible use for the robotic hand. Its accuracy supports the success of the hand from structure to movement to data collection. The random forest machine learning model trained on the four objects identifies the object by recognizing the unique grip patterns.

A future application that relies on this base includes adaptive intelligence in both robotic hands' grasp that can be used to support advanced prosthetics, agriculture automation for picking vegetables, cooking robots in preparing food, and laboratory handling of fragile glass objects. Another application is the use of this anthropomorphic hand in aiding low cost computer vision, as traditional computer vision methods like Lidar fail to account for the texture and compressibility of certain materials and geometries, while tactile sensors are able to account for these variables[22,23,24].

If the future applications mentioned above are met, the hand may be used in various ways and in multiple fields. The hand may be used as prosthetics to replace arms and hands of people who have lost them, by imitating a real hand and giving the person a way to touch and hold onto objects. The robotic hands may also be used in agricultural settings. With the hand's object detection, it can determine various types of fruits and vegetables to pick. In manufacturing, robotic hands can also use object detection to complete tasks needed to create items. For example, it can continuously repeat tasks on an assembly line.

## Conclusion

This hand exemplifies the applicability of low-cost tactile sensing in a 3D printed anthropomorphic robotic hand. Combined with a random forest object detection model, the tactile sensors were able to consistently detect individual objects from four base cases with accuracy ranging from 97% to 100%. The model had an overall accuracy of 98.1%, supporting the viability of this technology as a future aid to adaptive grasp and computer vision assistance, especially since it is significantly cheaper than existing anthropomorphic tactile sensing systems. Future research should aim to expand on the application of tactile sensing data, using it to support more active technical improvements like adaptive grasp or computer vision. Future research can also fully explore the viability of the suggested bioinspired electronic-skin finger ridge system, which could further enhance the accuracy and stability of the sensors in this particular low-cost hand [13,11].

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